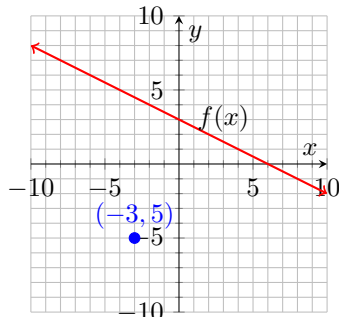


Plotting point on the graph and reading points from the graph

1. Given a point (x, y) count x *horizontally* in the direction of the sign and count y *vertically* in the direction of the sign.

Find the x and y intercepts from a graph. (See the midterm review for examples)

1. Find all the times the graph intersects the x -axis. These are your x -intercepts.
2. Find all the times the graph intersects the y -axis. These are your y -intercepts.



ex.

Solve an equation. ex. $3(x + 8) - 2 = -(x + 3) - 4x$.

1. Simplify, distribute, and then simplify. Note that $-(x + 3) = -1(x + 3)$.

$$\begin{aligned} 3x + 24 - 2 &= -x - 3 \\ 3x + 22 &= -x - 3 \end{aligned}$$

2. Combine like terms and solve, i.e., put " x 's" on one side and numbers on the other..

$$\begin{aligned} 4x &= -25 \\ x &= \frac{-25}{4} \end{aligned}$$

Solve an equation with fractions. ex. $\frac{2x}{6} + 2 = \frac{3x}{9}$

1. Multiply by the Lowest Common Denominator (LCD). In this case, 18. Note BOTH terms on the left got multiplied by 18.

$$\begin{aligned} 18\left(\frac{2x}{6} + 2\right) &= \left(\frac{x}{9}\right) 18 \\ \frac{36x}{6} + 36 &= \frac{18x}{9} \\ 6x + 36 &= 2x \end{aligned}$$

2. Now just solve.

$$\begin{aligned} 4x &= -36 \\ x &= -\frac{36}{4} = -9 \end{aligned}$$

Write the value(s) that make the denominator(s) zero. Then solve an equation. ex. $\frac{5}{x-2} +$

$$\frac{2}{x+7} = \frac{10}{(x+7)(x-2)}$$

1. First note that $x \neq 2, -7$.

2. Multiply by the LCD. In this case $(x - 2)(x + 7)$.

$$\begin{aligned}(x + 7)(x - 2) \left(\frac{5}{x - 2} + \frac{2}{x + 7} \right) &= \left(\frac{10}{(x + 7)(x - 2)} \right) (x + 7)(x - 2) \\ 5(x + 7) + 2(x - 2) &= 10\end{aligned}$$

3. Now solve.

$$\begin{aligned}5(x + 7) + 2(x - 2) &= 10 \\ 7x + 31 &= 10 \\ x &= -3\end{aligned}$$

Solve a formula for a specified variable. ex. $P = 2L + 2W$ for L .

1. Just solve like you do for an x , and treat the other letters like numbers.

$$\begin{aligned}P &= 2L + 2W \\ P - 2W &= 2L \\ \frac{P - 2W}{2} &= L\end{aligned}$$

2. Note: The 2 divides BOTH factors, and the you DO NOT cancel anything in $2W$.

Solve using the square root property. ex. $(x - 4)^2 + 1 = 2$.

1. Get the squared part by itself:

$$(x - 4)^2 = 1$$

2. Use a square root; Don't forget the $\pm$.

$$\begin{aligned}\sqrt{(x - 4)^2} &= \pm\sqrt{1} \\ x - 4 &= \pm 1\end{aligned}$$

3. Solve.

$$\begin{aligned}x &= \pm 1 + 4 \\ x = 5 \quad \text{and} \quad x &= 3\end{aligned}$$

Solve by factoring. ex. $3x^2 = -3x + 6$

1. Set the equation equal to 0.

$$3x^2 + x - 2 = 0$$

2. Factor any common multiples.

$$3(x^2 + x - 2) = 0$$

3. Factor and solve.

$$\begin{aligned}3(x + 2)(x - 1) &= 0 \\ x = -2 \quad \text{and} \quad x &= 1\end{aligned}$$

Solve the equation using the quadratic formula. ex. $3x^2 = -2x + 2$.

1. Set the equation equal to 0.

$$3x^2 + 2x - 2 = 0$$

2. Identify the a, b, c in $ax^2 + bx + c = 0$

$$a = 3, b = 2, c = -2$$

3. Use the quadratic formula, $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$, and simplify.

$$x = \frac{-2 \pm \sqrt{(2)^2 - 4(3)(-2)}}{2(3)} = \frac{-2 \pm \sqrt{28}}{6} = \frac{-1 \pm \sqrt{7}}{3}$$

Solve the equation by completing the square. ex. $x^2 = -4x + 2$.

1. Set the equation equal to “ c ”, in this case 2, and divide by the “ a ”.

$$x^2 + 4x = 2$$

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2. Add $\left(\frac{b}{2}\right)^2$ to both sides; in our example $b = 4$ so add $\left(\frac{4}{2}\right)^2 = (2)^2 = 4$.

$$x^2 + 4x + 4 = 2 + 4$$

3. Factor, and solve (note it will always be $(x + \frac{b}{2})^2$)

$$(x + 2)^2 = 6$$

$$x + 2 = \pm \sqrt{6}$$

$$x = -2 \pm \sqrt{6}$$

Use the 5 step strategy for solving the word problem. When 3 times the number is subtracted 7 times the number, the result is 44.

1. Summarize: “3 times a number” means, $3x$. “7 times a number”, means $7x$. “Subtracted from” means, $7x - 3x$. “Result is 44” means, $= 44$.
2. Write an equation: $7x - 3x = 44$
3. Solve: $x = 11$.
4. Check: $7(11) - 3(11) = 44\checkmark$
5. State your answer as a sentence (this isn’t applicable here).

Solve the word problem. ex. A car rental place charges \$300 a week to rent a car plus an additional \$0.50 per mile. How many miles can you drive in a week for \$400.

1. Write as an equation.

$$300 + 0.50x = 400$$

2. Solve.

$$\begin{aligned} 0.50x &= 100 \\ x &= \frac{100}{0.50} = 200 \text{ miles} \end{aligned}$$